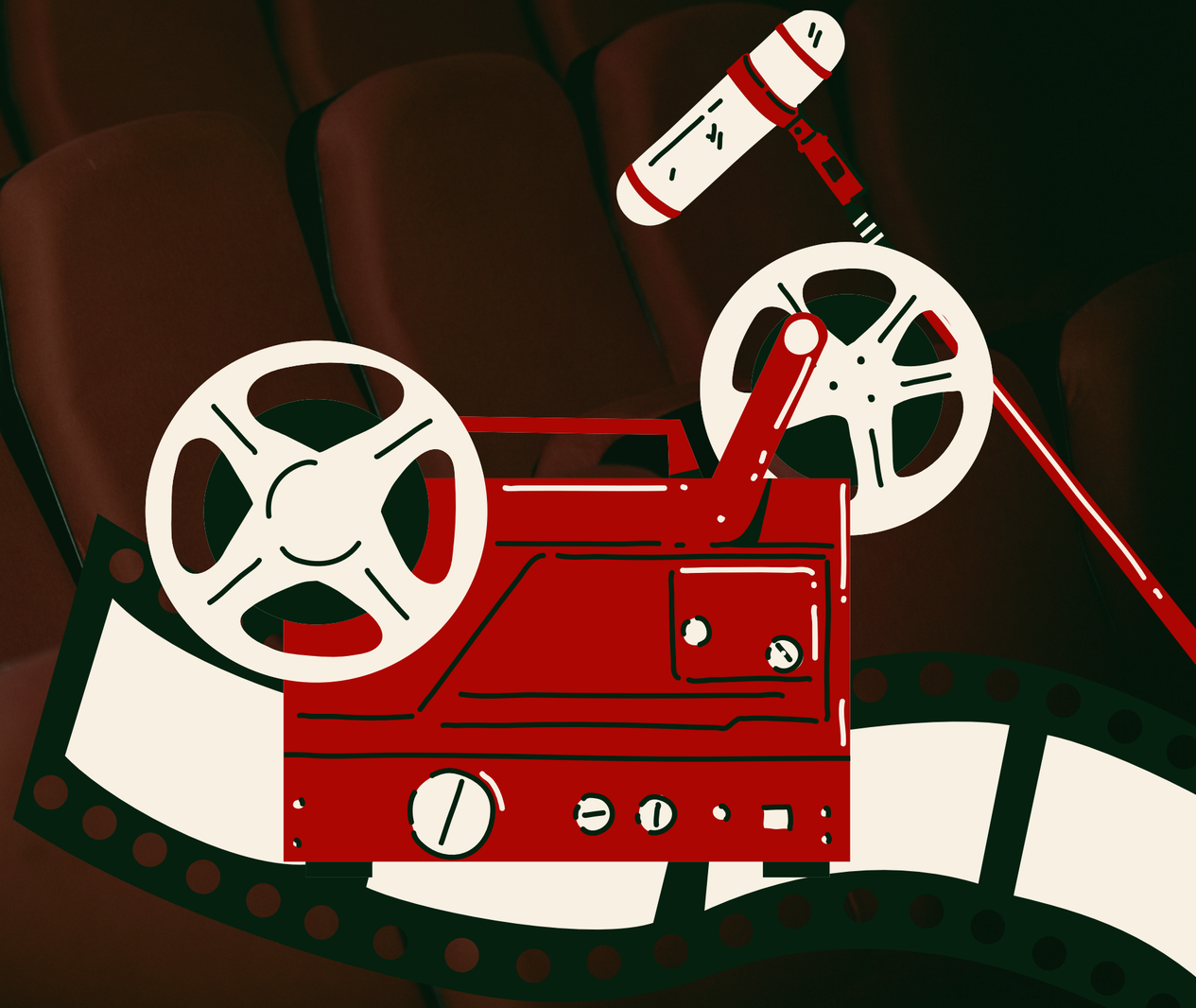
Decoding the Sakila Dataset with ppt

SAKILA ENTERTAINMENT



Presented By: **chandrakant kishor**

- The canonical sample database for MySQL.
- Simulates a complete DVD rental business ecosystem.
- Contains over 47,000 records across 16 interconnected tables.

Visual Suggestion

An icon of a database or a classic video rental store storefront.



Q1. Find the total number of films available in the database.

QUERY

```
SELECT
    COUNT(*) AS total_films
FROM
    film;
```

ANS

	total_films
▶	1000



Q2 . Find the total number of payment records in the database.

QUERY

```
SELECT
    COUNT(*) AS total_payments
FROM
    payment;
```

ANS

Result Grid

	total_payments
▶	16044



Q3. Find the minimum and maximum payment amount made by customers.

QUERY

```
SELECT
    MIN(amount) AS lowest_payment,
    MAX(amount) AS highest_payment
FROM
    payment;
```

ANS

	lowest_payment	highest_payment
▶	0.00	11.99



Q4. Find how many films belong to each category.

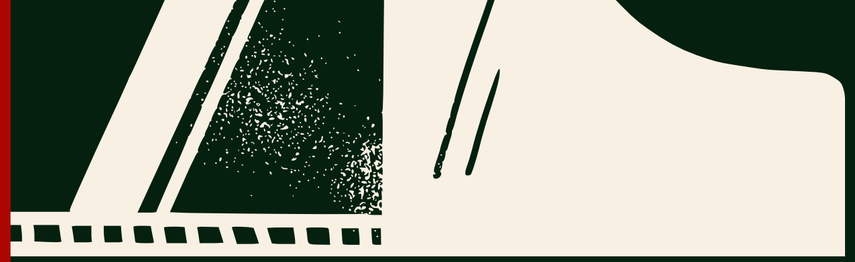
QUERY

```
SELECT
    c.name AS category_name,
    COUNT(fc.film_id) AS number_of_films
FROM
    category c
        JOIN
    film_category fc ON c.category_id = fc.category_id
GROUP BY c.name;
```

ANS

	total_films
▶	1000





Q5 Find categories that contain more than 60 films.

QUERY

```sql
SELECT
    c.name AS category_name,
    COUNT(fc.film_id) AS number_of_films
FROM
    category c
        JOIN
    film_category fc ON c.category_id = fc.category_id
GROUP BY c.name
HAVING COUNT(fc.film_id) > 60;
```

ANS

Result Grid | Filter Rows:

category_name	number_of_films
Action	64
Animation	66
Documentary	68
Drama	62
Family	69
Foreign	73
Games	61

Q6. Find actors who appeared in more than 20 films, sorted by number of films.

QUERY

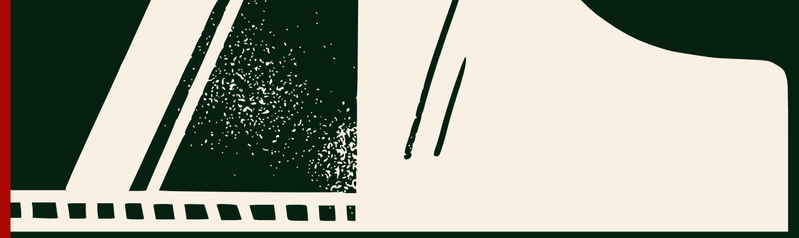
```
SELECT
    a.first_name,
    a.last_name,
    COUNT(fa.film_id) AS number_of_films
FROM
    actor a
        JOIN
    film_actor fa ON a.actor_id = fa.actor_id
GROUP BY a.actor_id , a.first_name , a.last_name
HAVING COUNT(fa.film_id) > 20
ORDER BY number_of_films DESC;
```

ANS

Result Grid | Filter Rows:

first_name	last_name	number_of_films
GINA	DEGENERES	42
WALTER	TORN	41
MARY	KEITEL	40
MATTHEW	CARREY	39
SANDRA	KILMER	37
SCARLETT	DAMON	36
UMA	WOOD	35





Q7. Display film title and the actor names who acted in those films.

QUERY

```
SELECT
    f.title AS film_title, a.first_name, a.last_name
FROM
    film f
        JOIN
    film_actor fa ON f.film_id = fa.film_id
        JOIN
    actor a ON fa.actor_id = a.actor_id;
```

ANS

film_title	first_name	last_name
CLASH FREDDY	VIVIEN	BERGEN
CRANES RESERVOIR	VIVIEN	BERGEN
DIRTY ACE	VIVIEN	BERGEN
DONNIE ALLEY	VIVIEN	BERGEN
DRIFTER COMMANDMENTS	VIVIEN	BERGEN
DRIVING POLISH	VIVIEN	BERGEN
DRUMS DYNAMITE	VIVIEN	BERGEN

Q8. List all films and their categories (even if some films have no category).

QUERY

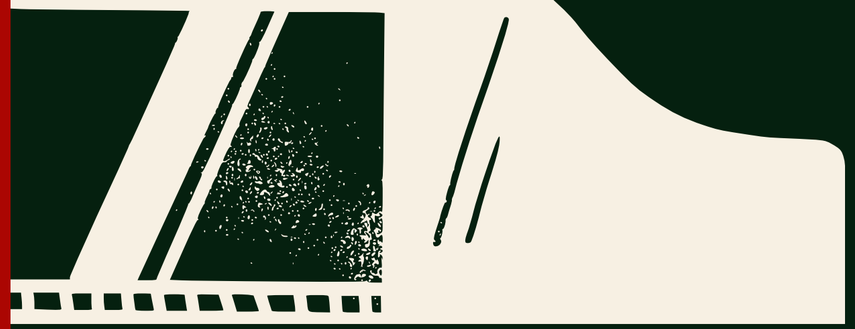
```
SELECT
    f.title AS film_title, c.name AS category_name
FROM
    film f
        LEFT JOIN
    film_category fc ON f.film_id = fc.film_id
        LEFT JOIN
    category c ON fc.category_id = c.category_id;
```

ANS

Result Grid | Filter Rows:

film_title	category_name
ACADEMY DINOSAUR	Documentary
ACE GOLDFINGER	Horror
ADAPTATION HOLES	Documentary
AFFAIR PREJUDICE	Horror
AFRICAN EGG	Family
AGENT TRUMAN	Foreign
AIRPLANE SIERRA	Comedy





Q9. Show all film categories and the films belonging to those categories.

QUERY

```sql
SELECT
    c.name AS category_name, f.title AS film_title
FROM
    category c
        LEFT JOIN
    film_category fc ON c.category_id = fc.category_id
        LEFT JOIN
    film f ON fc.film_id = f.film_id;
```

ANS

Result Grid | Filter Rows:

category_name	film_title
Action	AMADEUS HOLY
Action	AMERICAN CIRCUS
Action	ANTITRUST TOMATOES
Action	ARK RIDGEMONT
Action	BAREFOOT MANCHURIAN
Action	BERETS AGENT
Action	BRIDE INTRIGUE

10. Generate all combinations of customers and film categories.

QUERY

```
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    cat.name AS category_name
FROM
    customer c
        CROSS JOIN
    category cat;
```

ANS

Result Grid | Filter Rows: | Export:

customer_id	first_name	last_name	category_name
1	MARY	SMITH	Travel
1	MARY	SMITH	Sports
1	MARY	SMITH	Sci-Fi
1	MARY	SMITH	New
1	MARY	SMITH	Music
1	MARY	SMITH	Horror
1	MARY	SMITH	Games



Q11. The marketing team wants to launch a VIP loyalty program. Identify the top 5 customers who have spent the most money overall so they can be sent exclusive invitations.
Expected Output: First name, last name, email, and total amount spent.

ANS

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	email	total_spent
KARL	SEAL	KARL.SEAL@sakilacustomer.org	221.55
ELEANOR	HUNT	ELEANOR.HUNT@sakilacustomer.org	216.54
CLARA	SHAW	CLARA.SHAW@sakilacustomer.org	195.58
MARION	SNYDER	MARION.SNYDER@sakilacustomer.org	194.61
RHONDA	KENNEDY	RHONDA.KENNEDY@sakilacustomer.org	194.61

QUERY

```sql
SELECT
    c.first_name,
    c.last_name,
    c.email,
    SUM(p.amount) AS total_spent
FROM
    customer c
        JOIN
    payment p ON c.customer_id = p.customer_id
GROUP BY c.customer_id , c.first_name , c.last_name , c.email
ORDER BY total_spent DESC
LIMIT 5;
```

Q12. The purchasing manager needs to know which movie categories are the most lucrative to inform future DVD purchasing decisions. Find the total revenue generated by each film category and rank them from highest revenue to lowest.
Expected Output: Category name and total revenue.

QUERY

```
SELECT
    cat.name AS category_name, SUM(p.amount) AS total_revenue
FROM
    category cat
        JOIN
    film_category fc ON cat.category_id = fc.category_id
        JOIN
    inventory i ON fc.film_id = i.film_id
        JOIN
    rental r ON i.inventory_id = r.inventory_id
        JOIN
    payment p ON r.rental_id = p.rental_id
GROUP BY cat.name
ORDER BY total_revenue DESC;
```

ANS

Result Grid | Filter Rows:

category_name	total_revenue
Sports	5314.21
Sci-Fi	4756.98
Animation	4656.30
Drama	4587.39
Comedy	4383.58
Action	4375.85
New	4351.62



Q13. The promotions team is planning a movie marathon weekend and wants to feature
the most popular actors. Which actors have had their movies rented the most times? List the top 5 actors and their total rental counts.
Expected Output: Actor's first name, last name, and the total number of times their films have been rented.

QUERY

```sql
SELECT
    a.first_name,
    a.last_name,
    COUNT(r.rental_id) AS total_rentals
FROM
    actor a
        JOIN
    film_actor fa ON a.actor_id = fa.actor_id
        JOIN
    inventory i ON fa.film_id = i.film_id
        JOIN
    rental r ON i.inventory_id = r.inventory_id
GROUP BY a.actor_id , a.first_name , a.last_name
ORDER BY total_rentals DESC
LIMIT 5;
```



ANS

first_name	last_name	total_rentals
GINA	DEGENERES	753
MATTHEW	CARREY	678
MARY	KEITEL	674
ANGELA	WITHERSPOON	654
WALTER	TORN	640

Q14. Upper management wants a high-level performance comparison between the two physical store locations. Find the total number of rentals processed and the total revenue generated by each store. Expected Output: Store ID, total rentals, and total revenue.

QUERY

```
SELECT
    s.store_id,
    COUNT(r.rental_id) AS total_rentals,
    SUM(p.amount) AS total_revenue
FROM
    store s
        JOIN
    inventory i ON s.store_id = i.store_id
        JOIN
    rental r ON i.inventory_id = r.inventory_id
        JOIN
    payment p ON r.rental_id = p.rental_id
GROUP BY s.store_id
ORDER BY total_revenue DESC;
```

ANS

Result Grid | Filter Rows:

store_id	total_rentals	total_revenue
2	8121	33726.77
1	7923	33679.79

Q15. The store needs to clear up shelf space. Are there any movies in -- the inventory that have never been rented? -- Expected Output: Film title, inventory ID, and the store ID it belongs to.

QUERY

```
SELECT
    f.title, i.inventory_id, i.store_id
FROM
    film f
        JOIN
    inventory i ON f.film_id = i.film_id
        LEFT JOIN
    rental r ON i.inventory_id = r.inventory_id
WHERE
    r.rental_id IS NULL;
```

ANS

Result Grid | Filter Rows:

	title	inventory_id	store_id
▶	ACADEMY DINOSAUR	5	2



1: Database Architecture (The Schema)
Heading: Core Entity Relationships
Bullet Points:
Inventory: film, actor, category (1,000 films, 200 actors).
Business: store, staff, rental, payment (Over 16,000 rentals).
Clients: customer, address, city, country (599 active customers).
Visual Suggestion: A simplified flowchart or a 3-pillar graphic separating Inventory, Business, and Clients.

2: Key Business Questions
Heading: What Are We Trying to Solve?
Bullet Points:
Which movie categories drive the most revenue?
Who are our top-performing actors and most loyal customers?
How does store location impact overall sales?
Visual Suggestion: A large, stylized question mark overlapping a film strip.

3: Film & Inventory Analytics
Heading: What's on the Shelves?
Bullet Points:
Top Genres: Sports, Sci-Fi, and Animation lead the inventory count.
Rental Duration: Average movie rental spans 5 days.
Replacement Costs: Tracking average loss value for unreturned inventory.
Visual Suggestion: A bar chart graphic (you can build this using Canva's built-in chart tools) showing mock top categories.

4: Customer Insights
Heading: Knowing the Audience
Bullet Points:
599 active members driving over 16,000 transactions.
High retention rate among the top 10% of spenders.
Key Metric: Average revenue per user (ARPU) helps target loyalty programs.
Visual Suggestion: A user persona layout with a generic avatar next to key customer stats.

5: Revenue & Financial Health
Heading: Box Office Returns (Sales Data)
Bullet Points:
Total Revenue: Tracking cumulative payments across both main store branches.
Top Store: Store 1 slightly edges out Store 2 in total rentals.
Peak Days: Weekend rentals generate the highest volume of late fees and new rentals.
Visual Suggestion: A line graph showing an upward revenue trend or a large, bold callout number for Total Sales.

6: Conclusion & Recommendations
Heading: Actionable Next Steps
Bullet Points:
Optimize Stock: Increase inventory for top-grossing categories (Sports & Sci-Fi).
Loyalty Perks: Introduce rewards for top 50 customers to drive repeat business.
Dynamic Pricing: Consider weekend promotions to move lower-performing titles.
Visual Suggestion: A clean summary layout with checkmarks next to each action item.



Thank you so much



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